7. Expansion Startegies for Rag

(V1) → Expansion Startegies for Rag →

→ Understanding and interpreting user queries accurately is crucial for delivering precise and relevant responses.

→ Rag core has two essential techniques:
① Query parsing
② Entity recognition
→ are foundation technique used for improving the relevance and precision of rag.

① Query Parsing →
→ helps to disect the user input into meaningful components such as intents and keywords, which are critical ~~and~~ for accuracy & precision.

② Entity Recognition →
→ Entity Recognition identific ~~specifies~~ specific entities like people, places or organisations, adding context and focus to the search

→ spacy to identify entities and parse query provided by user.

For code → we need to download/install spacy
→ command →
> python -m spacy download en_core_web_sm.

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4) Efficient Use of context →

→ LLMs have limited context window
→ Rag intelligently retrieves only the most relevant chunks of information from a large corpus, feeding concise evidence to the model.

5) Transparency and Explainability →

→ Since Rag responses are based on retrieved documents, the system can cite sources or show supporting evidence.
→ This transparency builds user trust - crucial in enterprise, academic, and legal ~~academics~~ applications.

6) Scalability and Maintability →

→ Adding new information is as simple as updating the retrieved database, not retraining the model.

2) Representing Text Data Using Embeddings →

→ Some popular Embedding Techniques are →

1) Word2vec → captures semantic and syntatic meaning
2) Glove → captures global context
3) FastText → handles rare words.

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(8) Integrating knowledge Bases

V2 → Synonym Expansions and semantic Expansions

→ Semantic and synonym expansion are essential techniques that helps improve query retreival in rag system, particullary in technology focussed context.

→ Synonym expansion allows us to broaden a search by including related or alternate terms

→ For instance, a query about GPU's might include synonyms related terms like graphic card, parallel processors or accelerators.

→ These helps in capturing variations in phrases and improves relevancy by considering different ways of expressing similar ideas

→ ~~By~~ semantic expansion → uses embeddings to identify terms that are conceptually related even if they aren't direct synonyms.

→ For example → expanding a query on llm might yields terms like nlp ~~or~~ or transformer models, which are related conceptually to large language model and provide context alligned results that aren't restricted to exact matches.

V3 → Pseudo-Relevance feed back and Evaluation

→ Pseudo relevance feedback is a powerful iterative technique for refining queries in RAG systems.

→ It takes the initial retrieval results, assumes that the top relevant, and uses this feedback to adjust the original query.

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→ Core aspects of pseudo relevance feedback is query re-weighting, which identifies important terms in the initial results and increases their significance in the refined query.

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